

## Question 1

Most of the planets that have been discovered outside our solar system orbit G-type stars, like our Sun. In 2014, \_\_\_\_\_ researchers identified a planet orbiting KELT-9, a B-type star more than twice as massive and nearly twice as hot as the Sun. Called KELT-9b, it is one of the hottest planets ever discovered.

Which choice completes the text with the most logical transition?

- A. likewise,
- B. however,
- C. therefore,
- D. for example,

## Question 2

In 1873, Spanish scientist Santiago Ramón y Cajal observed that brain fibers have distinct boundaries with clear end points, a finding that went against earlier assumptions about the brain. \_\_\_\_\_ scientists had assumed that the brain was a continuous web of fused fibers, not a vast network of distinct, individual cells.

Which choice completes the text with the most logical transition?

- A. However,
- B. Previously,
- C. As a result,
- D. Likewise,

## Question 3

For years, biologists have experimented with using grime-eating bacteria rather than harsh chemicals to clean artworks, and results have been impressive overall. \_\_\_\_\_ these bacterial strains—which can metabolize centuries' worth of oil, glue, dirt, and other surface impurities without creating harmful byproducts—have proven more effective than traditional chemical cleaning methods.

Which choice completes the text with the most logical transition?

- A. However,
- B. In many cases,
- C. As a result,
- D. Additionally,

## Question 4

Alexander Lawrence Posey (1873–1908) varied his focus and tone depending on the genre in which he was writing. In his poetry, he used heartfelt language to evoke the beauty and peacefulness of his natural surroundings; in his journalism, \_\_\_\_\_ he employed humor and satire to comment on political issues affecting his Muskogee Creek community.

Which choice completes the text with the most logical transition?

- A. that is,
- B. granted,
- C. similarly,
- D. by contrast,

## Question 5

Because an achiral molecule is symmetrical, flipping it yields a structurally identical molecule. A flipped chiral molecule, \_\_\_\_\_ can be compared to a glove that has been turned inside out: it produces a structurally inverted molecule rather than an identical one.

Which choice completes the text with the most logical transition?

- A. in other words,
- B. by contrast,
- C. for example,
- D. similarly,

## Question 6

Jhumpa Lahiri's story collection *Interpreter of Maladies* features multiple stories about romantic relationships. In "This Blessed House," newlyweds argue over whether to replace items left by the previous owners of their new home. \_\_\_\_\_ in "A Temporary Matter," a husband and wife attempt to rekindle their relationship during a four-night blackout.

Which choice completes the text with the most logical transition?

- A. Granted,
- B. For example,
- C. Likewise,
- D. Hence,

## Question 7

Establishing Coordinated Universal Time (UTC) is no easy task. Each month, readings of a single second from atomic clocks around the world are taken and sent to the International Bureau of Weights and Measures (BIPM) in France. \_\_\_\_\_ BIPM metrologists perform the meticulous work of assembling these minutely disparate readings into a globally shared time standard.

Which choice completes the text with the most logical transition?

- A. There,
- B. In particular,
- C. For example,
- D. Conversely,

## Question 8

The chemical trimethylamine N-oxide not only gives fish their fishy smell but also protects them from crushing hydrostatic pressure in deep waters. Trimethylamine N-oxide strengthens the bonds between water molecules in a fish's body. \_\_\_\_\_ these water molecules maintain their linked structure at extreme depths, thus preventing pressure-related damage.

Which choice completes the text with the most logical transition?

- A. Nevertheless,
- B. As a result,
- C. However,
- D. For instance,

## Question 9

\_\_\_\_\_ A 2022 study by researchers Hala Altamimi and Qiaozhen Liu investigated the relationship between nonprofit arts organizations' spending and performance. \_\_\_\_\_ the researchers examined the correlation between how much 22,328 US arts nonprofits spent on overhead—operational costs such as equipment and fundraising—and how many people attended their events (a measure of overall success).

Which choice completes the text with the most logical transition?

- A. Thus,
- B. In addition,
- C. By comparison,
- D. Specifically,

## Question 10

Scientists long debated the origins of chondrules, tiny glass beads that formed in meteors billions of years ago. For decades, different theories were proposed, from lightning strikes to powerful rock collisions, but none had sufficient evidentiary support. \_\_\_\_\_ scientists found strong evidence that chondrules were formed by shock waves in nearby nebulae.

Which choice completes the text with the most logical transition?

- A. For example,
- B. Therefore,
- C. Similarly,
- D. Finally,

## Question 11

When, in the 1800s, geologists first realized that much of Earth had once been covered by great sheets of ice, some theorized that the phenomenon was cyclical, occurring at regular intervals. Each Ice Age is so destructive, though, that it largely erases the geological evidence of its predecessor. \_\_\_\_\_ geologists were unable to confirm the theory of cyclical Ice Ages until the 1960s.

Which choice completes the text with the most logical transition?

- A. Hence,
- B. Moreover,
- C. Nevertheless,
- D. Next,

## Question 12

In a 2022 analysis of 200 terms, researchers found a broad pattern of valence-dependent mutation for which negative words saw a faster rate of cognate replacement—\_\_\_\_\_ the rate at which a word will be replaced over time with a noncognate form. Adjectives (e.g., "afraid") saw the largest effect; nouns (e.g., "attack"), meanwhile, saw the smallest.

Which choice completes the text with the most logical transition?

- A. for example,
- B. likewise,
- C. in addition,
- D. that is,

## Question 13

With their distinctive cone shapes and steeply sloping sides, the volcanoes Hverfjall (Iceland) and Toliman (Guatemala) may look similar from afar. Tehnuka Ilanko and other volcanologists, \_\_\_\_\_ can tell by how each was formed that Hverfjall is a cinder cone volcano, while Toliman is a composite volcano.

Which choice completes the text with the most logical transition?

- A. for example,
- B. in addition,
- C. therefore,
- D. though,

**Question 14** Voting members of the 2002 Latin Grammys were impressed by Banda Cuisillos's album *Puras Rancheras Con Cuisillos* and its contribution to the banda genre, a form of regional Mexican music featuring large ensembles of wind instruments and drums that first developed in southern and central Mexico in the mid-nineteenth century. \_\_\_\_\_ they awarded the group the Best Banda Album award.

Which choice completes the text with the most logical transition?

- A. In contrast,
- B. Meanwhile,
- C. Nevertheless,
- D. Accordingly,

**Question 15** Scientists were able to isolate a relatively pure sample of selenium in 1817, the same year they first discovered the element's existence. \_\_\_\_\_ the isolation process took longer for molybdenum, which was isolated in its pure form three years after scientists first discovered it.

Which choice completes the text with the most logical transition?

- A. By contrast,
- B. Thus,
- C. Similarly,
- D. For instance,